

$$(1) \lim_{x \rightarrow 1} \frac{2x^2 - x - 1}{x - 1} \Rightarrow \frac{(2x+1)(x-1)}{\cancel{x-1}} \Rightarrow \lim_{x \rightarrow 1} 2x+1 = 3 //$$

$$(2) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x} \Rightarrow \frac{(x+2)(x-2)}{x(x-2)} \Rightarrow \frac{x+2}{x} \Rightarrow \lim_{x \rightarrow 2} \frac{4}{2} = 2 //$$

$$(3) \lim_{x \rightarrow 2} \frac{x^2 + 2x - 3}{4x - 3} \Rightarrow \frac{4+4-3}{8-3} \Rightarrow \frac{5}{5} \Rightarrow \lim_{x \rightarrow 2} 1 //$$

$$(4) \lim_{x \rightarrow 1} \left(\frac{2x^2 - x + 1}{3x - 2} \right)^2 \Rightarrow \left(\frac{2-1+1}{3-2} \right)^2 \Rightarrow \lim_{x \rightarrow 1} 4 //$$

$$(5) \lim_{x \rightarrow 2} \left(\frac{x^3 + 2x^2 - 3x + 2}{x^2 + 4x + 3} \right)^{\frac{1}{3}} \Rightarrow \left(\frac{8+8-6+2}{4+8+3} \right)^{\frac{1}{3}} \Rightarrow \left(\frac{12}{15} \right)^{\frac{1}{3}} \Rightarrow \left(\frac{4}{5} \right)^{\frac{1}{3}} \Rightarrow \lim_{x \rightarrow 2} \left(\frac{4}{5} \right)^{\frac{1}{3}}$$

$$(6) \lim_{x \rightarrow 2} \frac{3x+2}{x^2-6x+5} \Rightarrow \frac{6+2}{4-12+5} \Rightarrow \frac{8}{-3} \Rightarrow \lim_{x \rightarrow 2} -\frac{8}{3} //$$

$$(7) \lim_{x \rightarrow 2} \left(\frac{7x-4-5}{-4+6+4} \right)^3 \Rightarrow \left(\frac{3}{6} \right)^3 \Rightarrow \frac{1}{8} //$$

$$(8) \lim_{x \rightarrow -1} \left(\frac{2x^2 + 3x - 9}{5x - 4} \right)^{\frac{1}{2}} \Rightarrow \left(\frac{2-3-9}{-5-4} \right)^{\frac{1}{2}} \Rightarrow \left(\frac{-5}{-9} \right)^{\frac{1}{2}} \Rightarrow \frac{\sqrt{5}}{3} \Rightarrow \lim_{x \rightarrow -1} \frac{\sqrt{5}}{3} //$$

$$(9) \lim_{x \rightarrow 2} \frac{(2x^2 + 3x + 1)^{\frac{1}{2}}}{6 - 4x} \Rightarrow \frac{(16)^{\frac{1}{2}}}{6-8} \Rightarrow \frac{4}{-2} \Rightarrow \lim_{x \rightarrow 2} -2 //$$

$$(10) \lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^2 - x - 6} \Rightarrow \frac{(x-1)(x-3)}{(x+2)(x-3)} \Rightarrow \frac{x-1}{x+2} \Rightarrow \lim_{x \rightarrow 3} \frac{2}{5} //$$

$$(11) \lim_{x \rightarrow \frac{1}{2}} \frac{(2x-1)(x+3)}{(2x-1)(x-2)} \Rightarrow \frac{(\frac{1}{2}+3)}{(\frac{1}{2}-2)} \Rightarrow -\frac{7}{3} //$$

$$(13) \lim_{x \rightarrow 2} \frac{x^4 - 16}{-x^3 + 8} \Rightarrow \frac{(x+2x^2+4x+8)(x-2)}{-(x^3-2x^2-4)(x-2)} \Rightarrow \frac{8+8+8+8}{-4-4-4} \Rightarrow -\frac{8}{3} //$$

$$(14) \lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} \Rightarrow \frac{(x^2+x+1)(x-1)}{(x+1)(x-1)} \Rightarrow \frac{3}{2} //$$

$$(15) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \Rightarrow \frac{(x+1)(x-1)}{(x-1)} \Rightarrow x+1 = 2 //$$

$$(16) \lim_{x \rightarrow -2} \frac{4 - x^2}{2 + x} \Rightarrow \frac{(x+2)(2-x)}{(x+2)} \Rightarrow 2-x = 4 //$$

$$(19) \lim_{x \rightarrow -3} \Rightarrow f(-3) = \frac{12x+3}{(x+3)} \Rightarrow -3 //$$

* importante analisar o funcionamento da limite se querendo de -3.

$$(20) \lim_{x \rightarrow 1} \frac{2x^3 + x^2 - 4x + 1}{x^3 - 3x^2 + 5x - 3} \Rightarrow \frac{(2x^2 + 3x - 1)(x-1)}{(x^2 - 2x + 3)(x-1)} \Rightarrow \lim_{x \rightarrow 1} \frac{2x^2 + 3x - 1}{x^2 - 2x + 3} \Rightarrow \lim_{x \rightarrow 1} \frac{9}{2} = 4.5 //$$

$$(21) \lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 6x - 4}{x^3 - 4x^2 + 8x - 5} \Rightarrow \frac{(x^2 - 2x + 4)(x-1)}{(x^2 - 3x + 5)(x-1)} \Rightarrow \lim_{x \rightarrow 1} \frac{x^2 - 2x + 4}{x^2 - 3x + 5} = 1 //$$

$$(22) \lim_{x \rightarrow 2} \frac{x^4 - 10x + 4}{x^2 - 2x^2} \Rightarrow \frac{(x^3 + 2x^2 + 4x - 2)(x-2)}{x^2(x-2)} \Rightarrow \lim_{x \rightarrow 2} \frac{x^3 + 2x^2 + 4x - 2}{x^2} = 16 //$$

$$(24) \lim_{x \rightarrow 2} \frac{2(1^2) - 3(1)^{-2}}{2-2} = \frac{0}{0} ? \quad \left| \lim_{x \rightarrow 2} \frac{(x-2)(2x+1)}{(x-2)} \Rightarrow \lim_{x \rightarrow 2} 5 //$$

$$(28) \lim_{x \rightarrow 0} \frac{0^2 - x^2}{0^3 + x^3} \Rightarrow \lim_{x \rightarrow 0} \frac{(-x+0)(x+0)}{(x^2-0x+0^2)(x+0)} = \frac{-x+0}{x^2-0x+0^2} \Rightarrow \lim_{x \rightarrow 0} \frac{-(-0)+0}{(-0)^2-0(-0)+(-0)^2} \Rightarrow \lim_{x \rightarrow 0} \frac{-0}{3 \cdot 0^2} = \frac{0}{0} //$$

$$(29) \lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} \Rightarrow \lim_{x \rightarrow 1} \frac{(x-1)(x^{m-1} + x^{m-2} + \dots + x + 1)}{(x-1)(x^{n-1} + x^{n-2} + \dots + x + 1)}$$

$$(30) \lim_{x \rightarrow 3} \frac{\sqrt{1+x} - 2}{x - 3} \Rightarrow \lim_{x \rightarrow 3} \frac{2-2}{3-3} \Rightarrow \lim_{x \rightarrow 3} \frac{0}{0} ? \quad \left| \lim_{x \rightarrow 3} \frac{\sqrt{1+x} - 2}{x - 3} \cdot \frac{\sqrt{1+x} + 2}{\sqrt{1+x} + 2} \Rightarrow \lim_{x \rightarrow 3} \frac{(x-3)}{(x-3)(\sqrt{1+x}+2)} \Rightarrow \lim_{x \rightarrow 3} \frac{1}{\sqrt{1+x}+2} \Rightarrow \lim_{x \rightarrow 3} \frac{1}{4} //$$

$$(31) \lim_{x \rightarrow 0} \frac{1 - \sqrt{1-x}}{x} \rightarrow \lim_{x \rightarrow 0} \frac{1 - \sqrt{1-x}}{0} = ? \quad \left| \lim_{x \rightarrow 0} \frac{1 - \sqrt{1-x}}{x} \cdot \frac{-1 - \sqrt{1-x}}{-1 - \sqrt{1-x}} \rightarrow \lim_{x \rightarrow 0} \frac{-x}{x \cdot (-1 - \sqrt{1-x})} \Rightarrow \lim_{x \rightarrow 0} \frac{1}{1 + \sqrt{1-x}} \Rightarrow \lim_{x \rightarrow 0} \frac{1}{2} \right\|$$

$$(32) \lim_{x \rightarrow 0} \frac{\sqrt{-x^2 - 2x + 1} - 1}{x} \rightarrow \lim_{x \rightarrow 0} \frac{\sqrt{-x^2 - 2x + 1} - 1}{x} \cdot \frac{\sqrt{-x^2 - 2x + 1} + 1}{\sqrt{-x^2 - 2x + 1} + 1} \rightarrow \lim_{x \rightarrow 0} \frac{x \cdot (x-2)}{x \cdot (\sqrt{-x^2 - 2x + 1} + 1)} \rightarrow \lim_{x \rightarrow 0} -1 //$$

$$(33) \lim_{x \rightarrow 0} \frac{x^2 - 0^2}{x - 0} \rightarrow \lim_{x \rightarrow 0} \frac{0^2 - 0^2}{0 - 0} = \frac{0}{0} = ? \quad \left| \lim_{x \rightarrow 0} \frac{x^2 - 0^2}{x - 0} \rightarrow \frac{(x+0)(x-0)}{(x-0)} \rightarrow \lim_{x \rightarrow 0} (x+0) \Rightarrow \lim_{x \rightarrow 0} 0 \right\|$$

$$(34) \lim_{x \rightarrow 1} \frac{x^n - 1}{x - 1}$$

$$(35) \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} = \frac{1-1}{1-1} = \frac{0}{0} = ? \quad \left| \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} \cdot \frac{\sqrt{x} + 1}{\sqrt{x} + 1} = \frac{x-1}{(x-1)(\sqrt{x}+1)} \Rightarrow \lim_{x \rightarrow 1} \frac{1}{\sqrt{x}+1} = \frac{1}{2} // \right.$$

$$(36) \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x - 1} = \frac{2-2}{1-1} = \frac{0}{0} = ? \quad \left| \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x - 1} \cdot \frac{\sqrt{x+3} + 2}{\sqrt{x+3} + 2} = \frac{(\sqrt{x+3})^2 - 2^2}{(x-1)(\sqrt{x+3}+2)} = \frac{x-1}{(x-1)(\sqrt{x+3}+2)} = \frac{1}{\sqrt{x+3}+2} \Rightarrow \lim_{x \rightarrow 1} \frac{1}{\sqrt{x+3}+2} = \frac{1}{4} //$$

$$(37) \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} = \frac{1-1}{0} = \frac{0}{0} = ? \quad \left| \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} \cdot \frac{\sqrt{1+x} + \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} = \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})} = \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \Rightarrow \lim_{x \rightarrow 0} 1 \right.$$

$$(38) \lim_{x \rightarrow 1} \frac{\sqrt{2x^2} - \sqrt{x+1}}{x - 1} = \frac{\sqrt{2} - \sqrt{2}}{1-1} = \frac{0}{0} = ? \quad \left| \lim_{x \rightarrow 1} \frac{\sqrt{2x^2} - \sqrt{x+1}}{x - 1} \cdot \frac{\sqrt{2x^2} + \sqrt{x+1}}{\sqrt{2x^2} + \sqrt{x+1}} = \frac{(x-1)}{(x-1)(\sqrt{2x^2} + \sqrt{x+1})} = \frac{1}{\sqrt{2x^2} + \sqrt{x+1}} \Rightarrow \lim_{x \rightarrow 1} \frac{1}{\sqrt{2x^2} + \sqrt{x+1}} = \frac{\sqrt{2}}{4} //$$

$$(39) \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x^2 - 3x + 2} \cdot \frac{\sqrt{x+3} + 2}{\sqrt{x+3} + 2} = \frac{(x-1)}{(x-2)(x-1)(\sqrt{x+3}+2)} = \frac{1}{(x-2)(\sqrt{x+3}+2)} \Rightarrow \lim_{x \rightarrow 1} \frac{1}{(x-2)(\sqrt{x+3}+2)} = -\frac{1}{4} //$$

$$(40) \lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}} \cdot \frac{\sqrt{x+2} + \sqrt{3x-2}}{\sqrt{x+2} + \sqrt{3x-2}} = \frac{(x-2)(x+2)}{(x-2)(\sqrt{x+2} + \sqrt{3x-2})} = \frac{(x+2)(\sqrt{x+2} + \sqrt{3x-2})}{-2} \Rightarrow \lim_{x \rightarrow 2} \frac{(x+2)(\sqrt{x+2} + \sqrt{3x-2})}{-2} = -8 //$$

$$(41) \lim_{x \rightarrow 2} \frac{\sqrt{3x-2} - 2}{\sqrt{4x+1} - 3} \rightarrow \lim_{x \rightarrow 2} \frac{0}{0} = ? \quad \left| \frac{(\sqrt{3x-2} - 2)(\sqrt{4x+1} + 3)}{(\sqrt{4x+1} - 3)(\sqrt{4x+1} + 3)} \rightarrow \frac{(\sqrt{3x-2} - 2)(\sqrt{4x+1} + 3)}{4(x-2)} \rightarrow \frac{3(\sqrt{4x+1} + 3)}{4(x-2)} \rightarrow \frac{3(\sqrt{4x+1} + 3)}{4(\sqrt{3x-2} + 2)} \right.$$

$$\rightarrow \lim_{x \rightarrow 2} \frac{3(\sqrt{4x+1} + 3)}{4(\sqrt{3x-2} + 2)} = \frac{3(\sqrt{4 \cdot 2 + 1} + 3)}{4(\sqrt{3 \cdot 2 - 2} + 2)} = \frac{3 \cdot 6}{4 \cdot 4} = \frac{9}{8} \Rightarrow \lim_{x \rightarrow 2} \frac{9}{8}$$

* Esqueça $x=2$

$$\frac{(\sqrt{3x-2} - 2)(\sqrt{4x+1} + 3)}{(\sqrt{4x+1} - 3)(\sqrt{4x+1} + 3)} \rightarrow \frac{3(\sqrt{4x+1} + 3)}{4(\sqrt{3x-2} + 2)}$$

* Foi necessário fazer esse trabalho para sair do limite

* Podemos observar que se dividirmos os dois primeiros em 4, o resultado será 1, ou seja, o valor original é o mesmo.

$$(42) \lim_{x \rightarrow 3} \frac{2 - \sqrt{x+1}}{x^2 - 9} = \frac{2-2}{9-9} = \frac{0}{0} = ? \quad \left| \lim_{x \rightarrow 3} \frac{(2 - \sqrt{x+1})(1 + \sqrt{x+1})}{(x^2 - 9)(1 + \sqrt{x+1})} = \frac{(x-3)}{(x-3)(x+3)(1 + \sqrt{x+1})} = \frac{1}{(x+3)(1 + \sqrt{x+1})} = \lim_{x \rightarrow 3} \frac{1}{24} //$$

$$(43) \lim_{x \rightarrow 1} \frac{3 - \sqrt{10-x}}{x^2 - 1} \cdot \frac{-\sqrt{10-x} - 3}{-\sqrt{10-x} - 3} = \frac{-(x+1)}{(x-1)(x+1)(-\sqrt{10-x}-3)} \rightarrow \frac{1}{(x-1)(-\sqrt{10-x}-3)} \rightarrow \lim_{x \rightarrow 1} \frac{1}{-12}$$

$$(44) \lim_{x \rightarrow 1} \frac{x^2 - 1}{\sqrt{2x^2} - 1} \cdot \frac{\sqrt{2x^2} + 1}{\sqrt{2x^2} + 1} = \frac{(x-1)(x+1)(\sqrt{2x^2} + 1)}{(x-1)(\sqrt{2x^2} + 1)} = \lim_{x \rightarrow 1} 1 //$$

$$(45) \lim_{x \rightarrow 4} \frac{(\sqrt{2x+1} - 3)(\sqrt{x-2} + \sqrt{2})}{(\sqrt{x-2} - \sqrt{2})(\sqrt{x-2} + \sqrt{2})} = \frac{(\sqrt{2x+1} - 3)(\sqrt{x-2} + \sqrt{2})}{x - 4} = \frac{2(x-4)(\sqrt{x-2} + \sqrt{2})}{(\sqrt{2x+1} + 3)(x-4)} \rightarrow \lim_{x \rightarrow 4} \frac{2\sqrt{2}}{3}$$

$$\frac{(\sqrt{2x+1} - 3)(\sqrt{2x+1} + 3)}{(\sqrt{2x+1} + 3)} = \frac{2x - 9}{\sqrt{2x+1} + 3} = \frac{2(x-4)}{\sqrt{2x+1} + 3}$$

$$(46) \lim_{x \rightarrow 2} \frac{x-2}{\sqrt[3]{3x-5}-1} \cdot \frac{(\sqrt[3]{3x-5})^2 + \sqrt[3]{3x-5} + 1}{(\sqrt[3]{3x-5})^2 + \sqrt[3]{3x-5} + 1} = \frac{(x-2) \cdot (\sqrt[3]{3x-5})^2 + \sqrt[3]{3x-5} + 1}{3x-6} \rightarrow \frac{(x-2) \cdot (\sqrt[3]{3x-5})^2 + \sqrt[3]{3x-5} + 1}{3(x-2)} = \frac{(\sqrt[3]{3x-5})^2 + \sqrt[3]{3x-5} + 1}{3} = \lim_{x \rightarrow 2} 1$$

$$o = \sqrt[3]{3x-5} \quad (o^3 + ob + b^3) \cdot (o-b)$$

$$b = 1 \quad \left\{ \left(\sqrt[3]{3x-5} \right)^2 + \sqrt[3]{3x-5} \cdot 1 + 1 \right\} \rightarrow \left(\sqrt[3]{3x-5} \right)^2 + \sqrt[3]{3x-5} + 1$$

* diferença entre eles

$$(47) \lim_{x \rightarrow -1} \frac{(x+1)}{\sqrt[3]{2x+3}-1} \cdot \frac{(\sqrt[3]{2x+3})^2 + \sqrt[3]{2x+3} \cdot 1 + 1}{(\sqrt[3]{2x+3})^2 + \sqrt[3]{2x+3} \cdot 1 + 1} = \frac{(x+1) \cdot (\sqrt[3]{2x+3})^2 + \sqrt[3]{2x+3} \cdot 1 + 1}{2 \cdot (x+1)} = \frac{(\sqrt[3]{2x+3})^2 + \sqrt[3]{2x+3} \cdot 1 + 1}{2} \Rightarrow \lim_{x \rightarrow -1} \frac{3}{2}$$

$$o = \sqrt[3]{2x+3} \quad b = 1 \quad (o^3 + ob + b^3) \cdot (o-b) \quad \left(\sqrt[3]{2x+3} \right)^2 + \sqrt[3]{2x+3} \cdot 1 + 1$$

$$(48) \lim_{x \rightarrow 0} \frac{\sqrt[3]{8-2x+x^2}-2}{x-x^2} \rightarrow \frac{\sqrt[3]{8-2x+x^2}-2}{x-x^2}$$

$$o = 8-2x+x^2 \quad b = -2 \quad o^3 - b^3 =$$

$$(49) \lim_{x \rightarrow 0} \frac{1-\sqrt[3]{1-x}}{1+\sqrt[3]{3x-1}}$$

$$(50) \lim_{x \rightarrow -1} \frac{\sqrt[3]{x}+1}{x+1} \cdot \frac{(\sqrt[3]{x})^2 - \sqrt[3]{x} \cdot 1 + 1}{(\sqrt[3]{x})^2 - \sqrt[3]{x} \cdot 1 + 1} = \frac{x+1}{[(\sqrt[3]{x})^2 - \sqrt[3]{x} \cdot 1 + 1] \cdot (x+1)} \rightarrow \frac{1}{[(\sqrt[3]{x})^2 - \sqrt[3]{x} \cdot 1 + 1]} \rightarrow \frac{2}{3}$$

$$o = \sqrt[3]{x} \quad b = 1 \quad (o+b)(o^2 - ob + b^2) = o^3 + b^3 \rightarrow (\sqrt[3]{x}+1) \cdot (\sqrt[3]{x})^2 - \sqrt[3]{x} \cdot 1 + 1$$

$$(51) \lim_{x \rightarrow 1} \frac{(\sqrt{x}-1)}{(\sqrt[3]{x}-1)} \rightarrow \frac{\frac{x-1}{\sqrt{x}+1}}{\frac{x-1}{\sqrt[3]{x}+1}} \rightarrow \frac{\sqrt[3]{x}+1}{\sqrt{x}+1} \rightarrow \lim_{x \rightarrow 1} \frac{3}{2}$$

Denominador: $o = \sqrt[3]{x} \quad b = 1 \quad o^3 - b^3 \rightarrow x-1 = (\sqrt[3]{x}-1) \cdot (\sqrt[3]{x}^2 + \sqrt[3]{x} \cdot 1 + 1) \rightarrow (\sqrt[3]{x}-1) = \frac{(x-1)}{(\sqrt[3]{x}^2 + \sqrt[3]{x} \cdot 1 + 1)}$

Numerador: $o = \sqrt{x} \quad b = 1 \quad o^2 - b^2 \rightarrow x-1 = (\sqrt{x}-1) \cdot (\sqrt{x}+1) \rightarrow \sqrt{x}-1 = \frac{x-1}{\sqrt{x}+1}$

$$(52) \lim_{x \rightarrow 64} \frac{\sqrt{x}-8}{\sqrt[3]{x}-4} \rightarrow \frac{\frac{x-64}{\sqrt{x}+8}}{\frac{x-64}{(\sqrt[3]{x})^2 + \sqrt[3]{x} \cdot 4 + 16}} \rightarrow \frac{(\sqrt[3]{x})^2 + \sqrt[3]{x} \cdot 4 + 16}{\sqrt{x}+8} \rightarrow \frac{16+16+16}{16} \rightarrow \frac{3 \cdot 16}{16} \rightarrow \lim_{x \rightarrow 64} 3$$

Denominador: $o = \sqrt[3]{x} \quad o^3 - b^3 = (o-b)(o^2 + ob + b^2)$

$$b = 4 \quad x-64 = (\sqrt[3]{x}-4) \cdot \textcircled{1} \rightarrow \sqrt[3]{x}-4 = \frac{x-64}{(\sqrt[3]{x})^2 + \sqrt[3]{x} \cdot 4 + 16}$$

Numerador: $o = \sqrt{x} \quad \frac{\sqrt{x}-8}{\sqrt{x}+8} = \frac{x-64}{\sqrt{x}+8}$ *basta usar esse tipo de artifício, pois não vai alterar meu valor final

$$(53) \lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1}-1}{x} \rightarrow \frac{(\sqrt[3]{x+1})^2 + (\sqrt[3]{x+1} \cdot 1) + 1}{1} \rightarrow \lim_{x \rightarrow 0} \frac{2}{3}$$

$$o = \sqrt[3]{x+1} \quad o^3 - b^3 = (o-b)(o^2 + ob + b^2) \quad \text{em resumo} \quad \sqrt[3]{x+1}-1 = \frac{x}{(\sqrt[3]{x+1})^2 + (\sqrt[3]{x+1} \cdot 1) + 1}$$

$$(54) \lim_{x \rightarrow 0} \frac{1-\sqrt[3]{1-x}}{1+\sqrt[3]{3x-1}} \rightarrow \frac{\frac{1-(1-x)^{1/3}}{1+(1-x)^{1/3}}}{\frac{1-(1-x)^{1/3}}{3[(1-x)^{2/3} + (1-x)^{1/3} + 1]}} \rightarrow \frac{3[(1-x)^{2/3} + (1-x)^{1/3} + 1]}{1+(1-x)^{1/3}} \rightarrow \lim_{x \rightarrow 0} \frac{2}{3}$$

Denominador: $o = 1 \quad 3x = 1 - (\sqrt[3]{1-x})^3 \rightarrow \sqrt[3]{1-x} = \frac{1-x}{(\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1}$

Numerador: $o = 1 \quad b = \sqrt[3]{1-x} \quad x = 1 - (\sqrt[3]{1-x})^3 \rightarrow \sqrt[3]{1-x} = \frac{x}{(\sqrt[3]{1-x})^2 + \sqrt[3]{1-x} + 1}$

$$(55) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{1 - \tan x} \rightarrow \frac{\sin x - \cos x}{1 - \frac{\sin x}{\cos x}} \rightarrow \frac{\sin x - \cos x}{\frac{\cos x - \sin x}{\cos x}} \rightarrow \frac{(\sin x - \cos) \cdot \cos x}{-1 \cdot (\sin x - \cos x)} \rightarrow -\cos x \therefore \lim_{x \rightarrow \frac{\pi}{4}} -\frac{\sqrt{2}}{2}$$

* multipliquei por (-1) para ficar igual ao numerador

$$(56) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^2 x} \rightarrow \frac{\frac{\sin x}{\cos x} - \sin x}{\sin^2 x} \rightarrow \frac{\frac{1}{\cos} - 1}{\sin x} \rightarrow \frac{\frac{1 - \cos x}{\cos x}}{\sin x} \rightarrow \frac{1 - \cos^2 x}{\sin x \cdot (1 + \cos x)} \rightarrow \frac{\sin^2 x}{(1 + \cos x) \cdot \sin x} \rightarrow \lim_{x \rightarrow 0} \frac{\sin x}{1 + \cos x} = 0$$

$\sin = 0$
 $\cos = 1$

$$\frac{1 - \cos x}{\sin x} \rightarrow \frac{1 - \cos x}{\sin x} \cdot \frac{1 + \cos x}{1 + \cos x} \rightarrow \frac{1 - \cos^2 x}{\sin x \cdot (1 + \cos x)}$$

$$(57) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos 2x}{\cos x - \sin x} \rightarrow \frac{\cos^2 x - \sin^2 x}{\cos x - \sin x} \rightarrow \frac{(\cos x - \sin x) \cdot (\cos x + \sin x)}{\cos x - \sin x} \rightarrow \lim_{x \rightarrow \frac{\pi}{4}} \cos x + \sin x = \sqrt{2}$$

$$\sin = \frac{\sqrt{2}}{2}$$

$$a^2 - b^2 \rightarrow (a-b) \cdot (a+b)$$

$$\cos = \frac{\sqrt{2}}{2}$$